PROJECT REPORT

STUDENT PERFORMANCE AI

Machine Learning Based Student Result Prediction

Submitted By

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ABSTRACT

- Student Performance AI is a machine-learning-based application designed to predict student results as PASS or FAIL based on selected student details. The project is developed using Python and the Decision Tree Classifier algorithm. Streamlit is used to create a simple and interactive web interface.

- The application uses features such as study hours, attendance, and previous marks to demonstrate the prediction process. The dataset is prepared using Pandas, and the machine learning model is trained using Scikit-learn. The trained model is integrated into the Streamlit application, allowing users to enter student details and view the predicted result.

- The main objective of this project is to understand the basic concepts of machine learning, data preparation, model training, prediction, and web application development. This application is intended for educational demonstration only. Since the model uses a small sample dataset, its predictions are not reliable guarantees of actual academic performance.

1.INTRODUCTION

- Machine learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. It is widely used in different fields, including education, healthcare, business, and technology.

- Student Performance AI is a machine-learning-based project developed to demonstrate how student-related information can be used to predict academic results. The application uses selected details such as study hours, attendance, and previous marks to predict whether a student may PASS or FAIL.

- The project uses Python for programming, Scikit-learn to build the machine-learning model, and Streamlit to develop the web interface. A Decision Tree Classifier is used to learn patterns from the sample dataset and generate predictions based on user input.

- The application provides a simple interface where users can enter student details and view the predicted result. This project helps demonstrate the basic workflow of machine learning, including data preparation, model training, prediction, and web application development.

- The project is intended for educational demonstration only. Since it uses a small sample dataset, its predictions should not be considered reliable guarantees of actual academic performance.

2. PROBLEM STATEMENT

Understanding how student-related information can be used in machine learning is an important learning objective. This project demonstrates how selected details, such as study hours, attendance, and previous marks, can be processed by a classification model to predict a student's result.

The aim is to develop a simple web application that accepts student details and displays a predicted PASS or FAIL result using a trained machine-learning model.

3. OBJECTIVES

The main objectives of this project are:

- To understand the basic concepts of machine learning and classification.
- To prepare a sample dataset containing student-related information.
- To divide the dataset into training and testing subsets.
- To train a Decision Tree Classifier for student result prediction.
- To develop a simple web interface using Streamlit.
- To display PASS or FAIL predictions based on user input.
- To understand the limitations of predictions made using a small sample dataset.

4. TECHNOLOGIES USED

The following technologies and libraries are used to develop the Student Performance AI project.

- **Python:** The main programming language used for model development and application logic.
- **Pandas:** Used to organize and handle the sample dataset.
- **NumPy:** Supports numerical operations.
- **Scikit-learn:** Used for splitting the dataset, training the Decision Tree Classifier, and evaluating predictions.
- **Matplotlib:** Used to create visualizations.
- **Seaborn:** Used to visualize the confusion matrix.
- **Joblib:** Used to save and load the trained machine-learning model.
- **Streamlit:** Used to create the interactive web application.
- **Jupyter Notebook:** Used to develop and test the machine-learning model.

5. DATASET DESCRIPTION

The project uses a small, manually created sample dataset for demonstration. The dataset contains three input features and one target column.

Input Features:

- **Study Hours:** The number of hours spent studying.
- **Attendance:** The student's attendance value.
- **Previous Marks:** The student's previous academic marks.

Target Column:

- **Result:** Represents the predicted class, where 0 indicates FAIL and 1 indicates PASS.

The dataset is used to train a Decision Tree Classifier. The trained model learns patterns from the sample records and uses them to predict the result for new inputs.

Note: The dataset is small and illustrative. Predictions are not reliable guarantees of actual academic performance.

6. SYSTEM WORKFLOW

The Student Performance AI project follows a series of steps to predict student results. First, a sample dataset containing study hours, attendance, and previous marks is prepared. The dataset is divided into input features and the target result.

Next, the dataset is split into training and testing data. A Decision Tree Classifier is trained using the training data. The trained model is then evaluated using suitable performance metrics. Finally, the model is connected to a Streamlit web application, where users enter student details and receive a predicted PASS or FAIL result.

Workflow: Data Preparation → Training → Testing → User Input → PASS/FAIL Prediction

7. IMPLEMENTATION

7.1 Data Preparation

The sample dataset is organized using Pandas. Study hours, attendance, and previous marks are used as input features, while the result column represents PASS or FAIL.

7.2 Model Training

The project uses the Decision Tree Classifier algorithm from Scikit-learn. The dataset is divided into training and testing subsets. The model learns patterns from the training data to predict the result for new inputs.

7.3 Model Evaluation

The model's performance can be examined using accuracy, a classification report, and a confusion matrix. These evaluation methods help demonstrate how the model predicts results on the testing data. However, the small sample dataset limits the reliability of the evaluation.

7.4 Web Application

The application is developed using Streamlit. It loads the saved machine-learning model and provides input fields for student details. When the user submits the information, the application displays the predicted result as PASS or FAIL.

8. RESULTS AND OUTPUT

- The Student Performance AI application successfully demonstrates student result prediction using a Decision Tree Classifier. The Streamlit website allows users to enter student details and displays the predicted result as PASS or FAIL.

- The project demonstrates the integration of a machine-learning model with a web application. However, the predictions are intended for educational demonstration only because the model uses a small sample dataset.

9. CONCLUSION

- Student Performance AI demonstrates the use of machine learning to predict student results based on selected input features. The project uses Python, Pandas, Scikit-learn, and Streamlit to develop a simple prediction system.

- Through this project, the basic concepts of data preparation, model training, testing, prediction, and web application development are demonstrated. In the future, the project could be improved by using a larger dataset and comparing different machine-learning algorithms.

10. REFERENCES

1. Python Documentation — https://docs.python.org/3/
2. Pandas Documentation — https://pandas.pydata.org/docs/
3. Scikit-learn Documentation — https://scikit-learn.org/stable/
4. Streamlit Documentation — https://docs.streamlit.io/